



Smart Restaurant Management System Using Generative AI

PRAKHAR SHARMA

Problem Definition and Requirement Analysis

- ▶ The modern restaurant industry faces significant operational inefficiencies that impact overall business performance, including:
- ▶ **Ineffective Menu Personalization:** Static menus that fail to adapt to customer preferences and emerging food trends.
- ▶ **Inventory Management Challenges:** Lack of real-time tracking leads to overstocking, wastage, and shortages.
- ▶ **Suboptimal Customer Engagement:** Limited personalization in promotions and loyalty programs reduces customer retention.
- ▶ **Event Planning Inefficiencies:** Restaurants struggle to optimize the scheduling and promotion of events.
- ▶ **Operational Overhead:** High dependency on manual decision-making for marketing and menu updates.

2. Objectives

- ▶ To address these issues, the Smart Restaurant Management System will leverage Generative AI to:
- ▶ **Automate and personalize** chef menu recommendations based on ingredient availability and customer preferences.
- ▶ Generate AI-driven promotional content for targeted customer engagement.
- ▶ Implement an advanced visual search and menu personalization system using image recognition.
- ▶ Provide automated event planning and optimization based on restaurant-specific metrics.
- ▶ Enhance real-time inventory tracking and predictive restocking using AI analytics.

Industry Research and Analysis

- ▶ **1. Market Trends and Challenges**
- ▶ **Personalization & AI Adoption:** The food industry is increasingly shifting towards AI-driven solutions for hyper-personalization, leading to increased revenue and customer satisfaction.
- ▶ **Inventory & Waste Management:** 70% of restaurant losses are due to inefficient inventory tracking and wasteful procurement.
- ▶ **Dynamic Pricing & Promotions:** AI-driven dynamic pricing models improve revenue optimization by adapting to demand fluctuations.

2. Key Pain Points Identified

- ▶ **Limited AI Integration in Traditional Restaurant Operations:** Most restaurants still rely on manual processes for menu planning, customer engagement, and promotions.
- ▶ **Lack of Real-Time Data Utilization:** Restaurants fail to leverage real-time customer behavior and inventory analytics.
- ▶ **Event Planning Complexity:** Inefficient handling of promotional campaigns and special events leads to underutilization of resources.

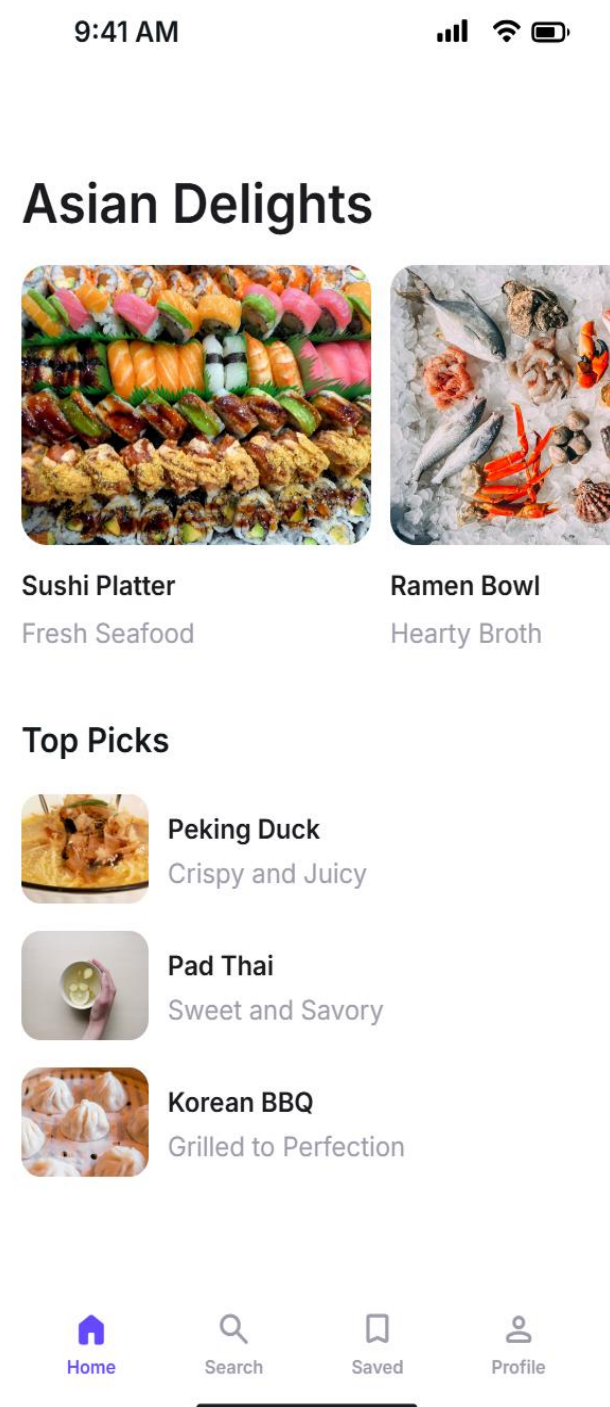
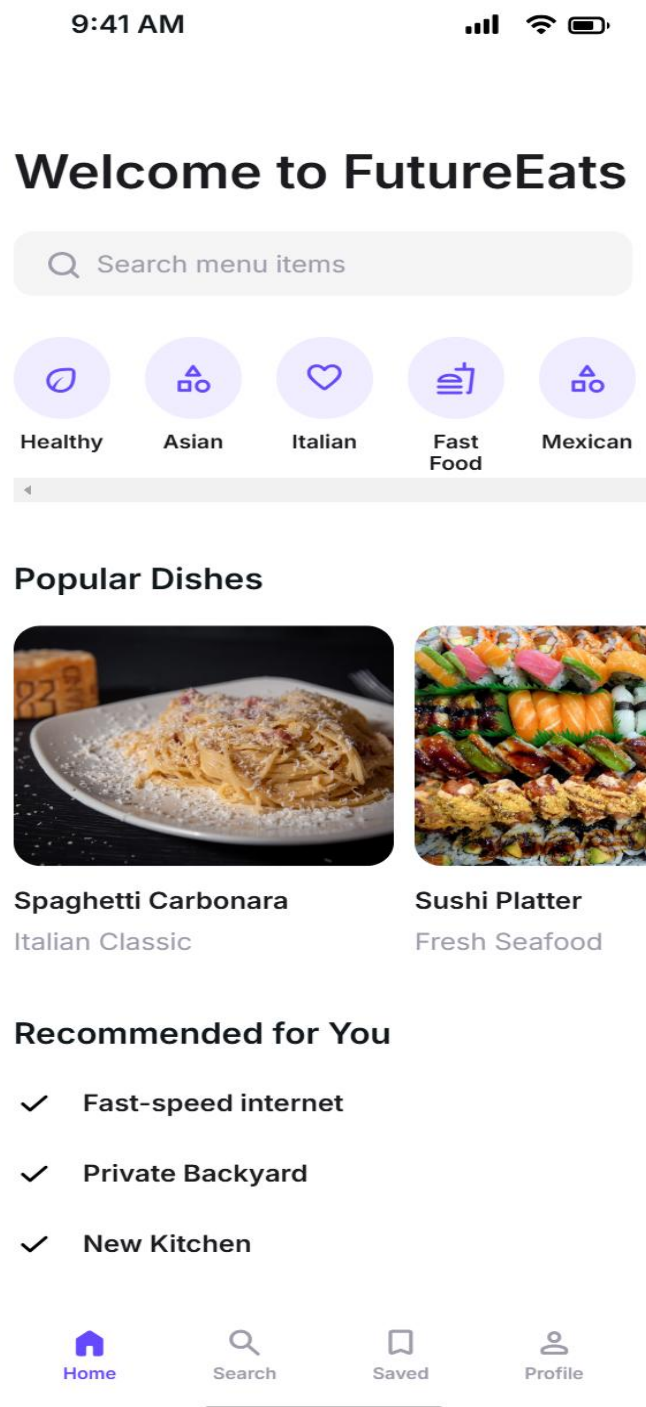
Feature Definition & Technical Specifications

- ▶ **1. AI-generated recipe Suggestions for Chefs**
- ▶ **Generative AI Model:** Trained on global cuisine datasets, dietary preferences, and seasonal ingredient availability.
- ▶ **Integration:** Seamless backend connectivity with inventory management systems for real-time ingredient status.
- ▶ **Personalization Algorithms:** Dynamic adaptation based on customer feedback and historical sales data.

2. Generative AI-Driven Promotions Creation

- ▶ **AI Content Generation:** Automatic generation of targeted promotional campaigns using NLP models.
- ▶ **Behavioral Analysis:** AI-based segmentation for personalized discounts, offers, and engagement campaigns.
- ▶ **Multi-Platform Deployment:** Auto-adaptation of promotions for email, social media, and in-app notifications.

USER INTERFACE



9:41 AM

Discover Dishes

Q

Search for dishes or cuisines

Popular Searches

Healthy

Asian

Italian

Fast Food

Mexican

Trending Now



Sushi Rolls
Fresh and Delicious



Vegan Tacos
Plant-based Deligh

9:41 AM

Fast Food Favorites



Cheeseburger
Juicy and Tasty



Pepperoni Pizza
Classic Favorite

Recommended



Loaded Nachos
Cheesy and Spicy



Hot Dogs
Grilled to Perfection



Onion Rings
Crispy and Golden

9:41 AM

Healthy Options



Quinoa Salad
Fresh and Nutritious



Avocado Toast
Healthy Breakfast

Top Picks



Kale Smoothie
Boost your energy



Vegan Bowl
Plant-based goodness



Chia Pudding
Perfect dessert

9:41 AM



Indian Cuisine
Explore the flavors of India

Popular Dishes



Biryani
Aromatic Rice Dish

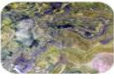


Paneer Butter Mas
Creamy Cottage Ch

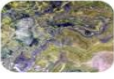
Recommended



Chole Bhature
Spicy Chickpeas



Rogan Josh
Tender Lamb Curry



Dosa
Crispy Rice Crepe



Tandoori Chicken
Grilled Spiced Chicken

Mexican Delights



Tacos Al Pastor
Spicy and Savory



Churros
Sweet Treat

Fast Food Favorites



Cheeseburger
Juicy and Tasty



Pepperoni Pizza
Classic Favorite

Recommended



Burrito Bowl
Hearty and Filling



Enchiladas
Rich and Flavorful



Guacamole
Fresh and Zesty

Recommended



Loaded Nachos
Cheesy and Spicy

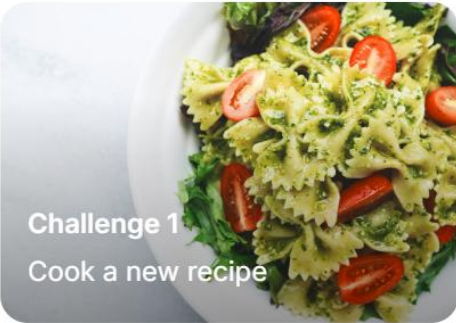


Hot Dogs
Grilled to Perfection



Onion Rings
Crispy and Golden

Daily Challenges



Challenge 1
Cook a new recipe



Challenge
Try a new

Earn Points

Complete tasks
Earn 50 points each

Share with friends
Earn 20 points

Daily login
Earn 10 points



Cook a New Recipe
Challenge 1

Challenge Details



Recipe Ideas
Explore new recipes to try



Healthy Options
Focus on nutritious meals



Asian Cuisine
Discover Asian flavors

Start Cooking

Share Challenge



Home



Search



Saved



Profile



Home



Search



Saved



Profile



Home



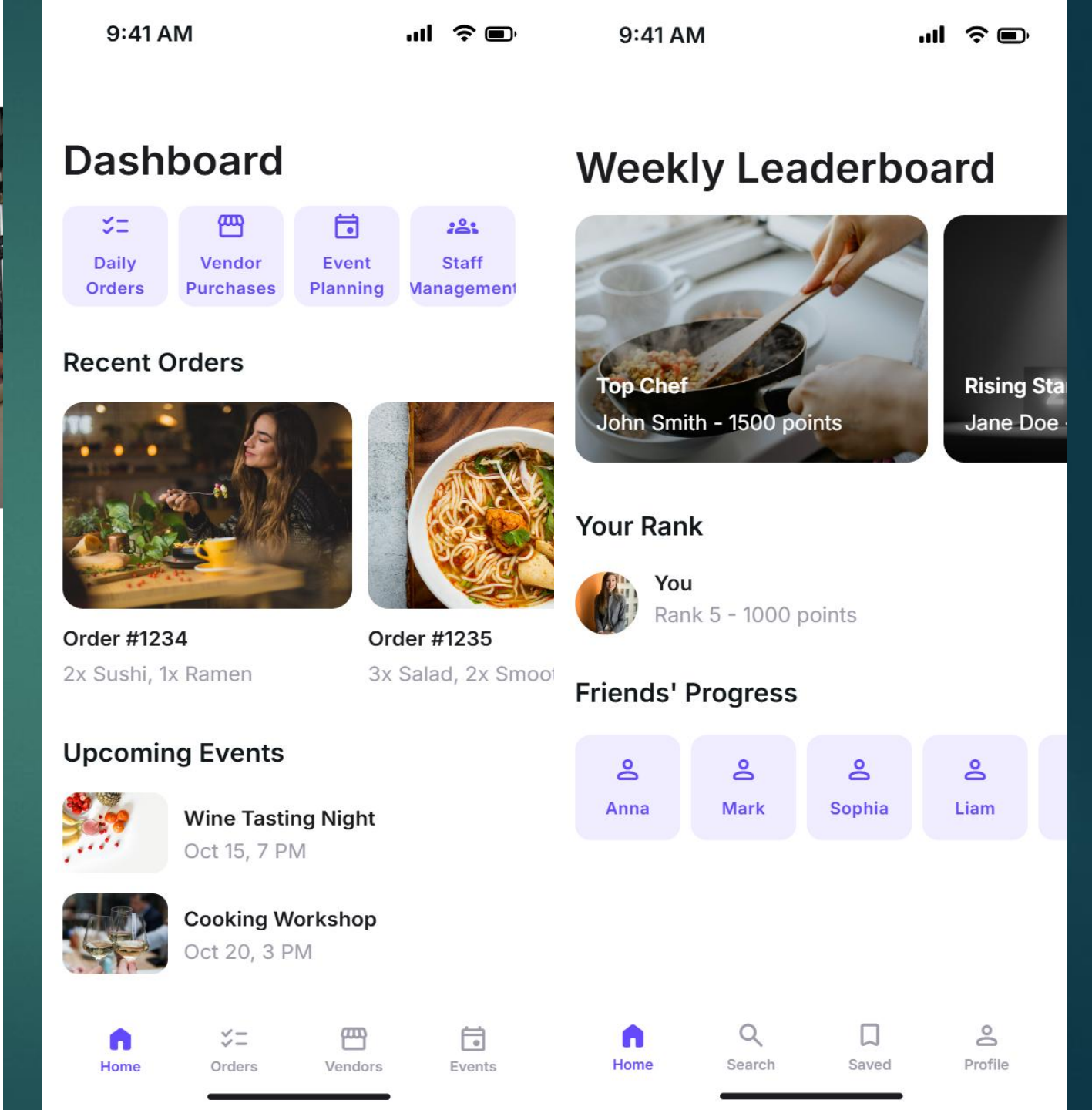
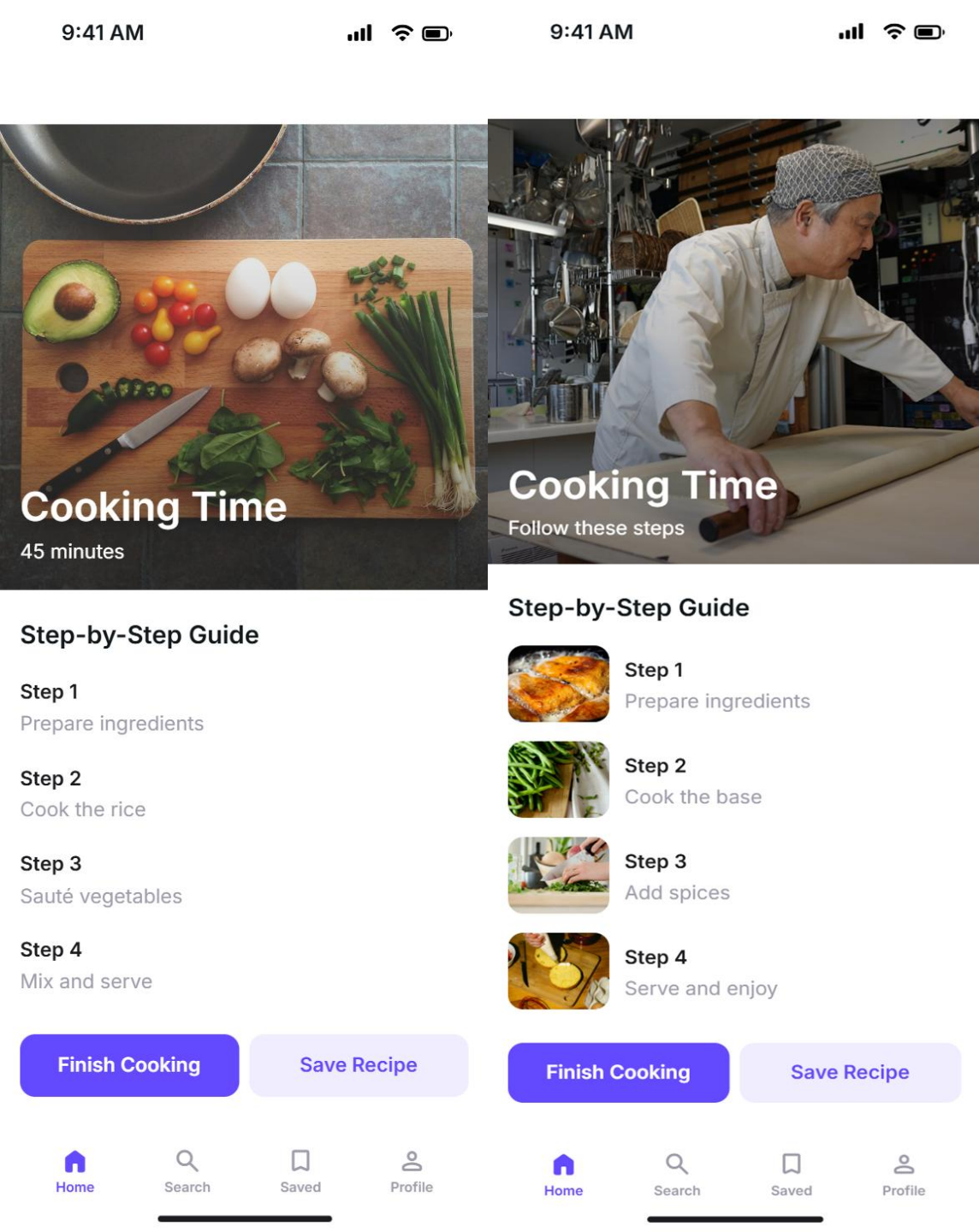
Search



Saved



Profile



Daily Orders

Search orders



Order #1234
2x Sushi, 1x Ramen



Order #1235
3x Salad, 2x Smoothie



Order #1236
1x Burger, 2x Fries

Add Order

View All



Home



Orders



Vendors



Events

Vendor Purchases

Search vendors

Recent Purchases



Order #5678
10x Tomatoes, 5x Lettuce



Order #5679
20x Chicken Breasts

Top Vendors



Fresh Farm Produce
Quality vegetables and fruits



Meat Masters
Premium meats and poultry



Home



Orders



Vendors



Events

Event Planning



Create Event



Manage Guests



Catering



Logistics

Upcoming Events



Wine Tasting Night
Oct 15, 7 PM



Cooking Workshop
Oct 20, 3 PM

Event Tools



Budgeting



Staffing



Marketing



Feedback



Home



Orders



Vendors



Events

Level Up!



Daily Challenges
Complete tasks to earn points



Weekly Challenges
Complete tasks to earn points

Your Progress

✓ Fast-speed internet



Home



Search

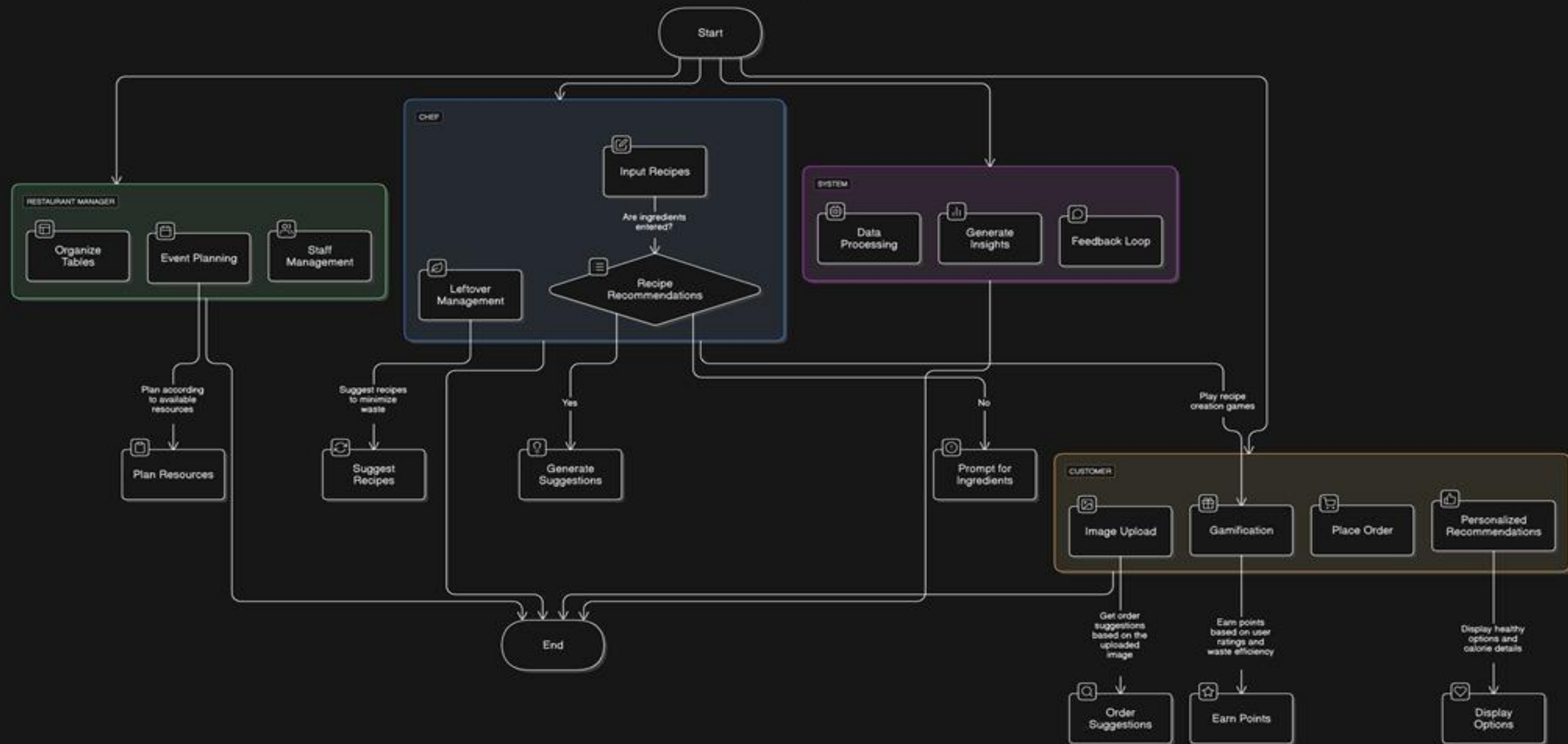


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Profile

Smart Restaurant Management App Flowchart



Present Features

Event Manager

Food Menu Recommendation

Leftover Management

Multimodels For Chef

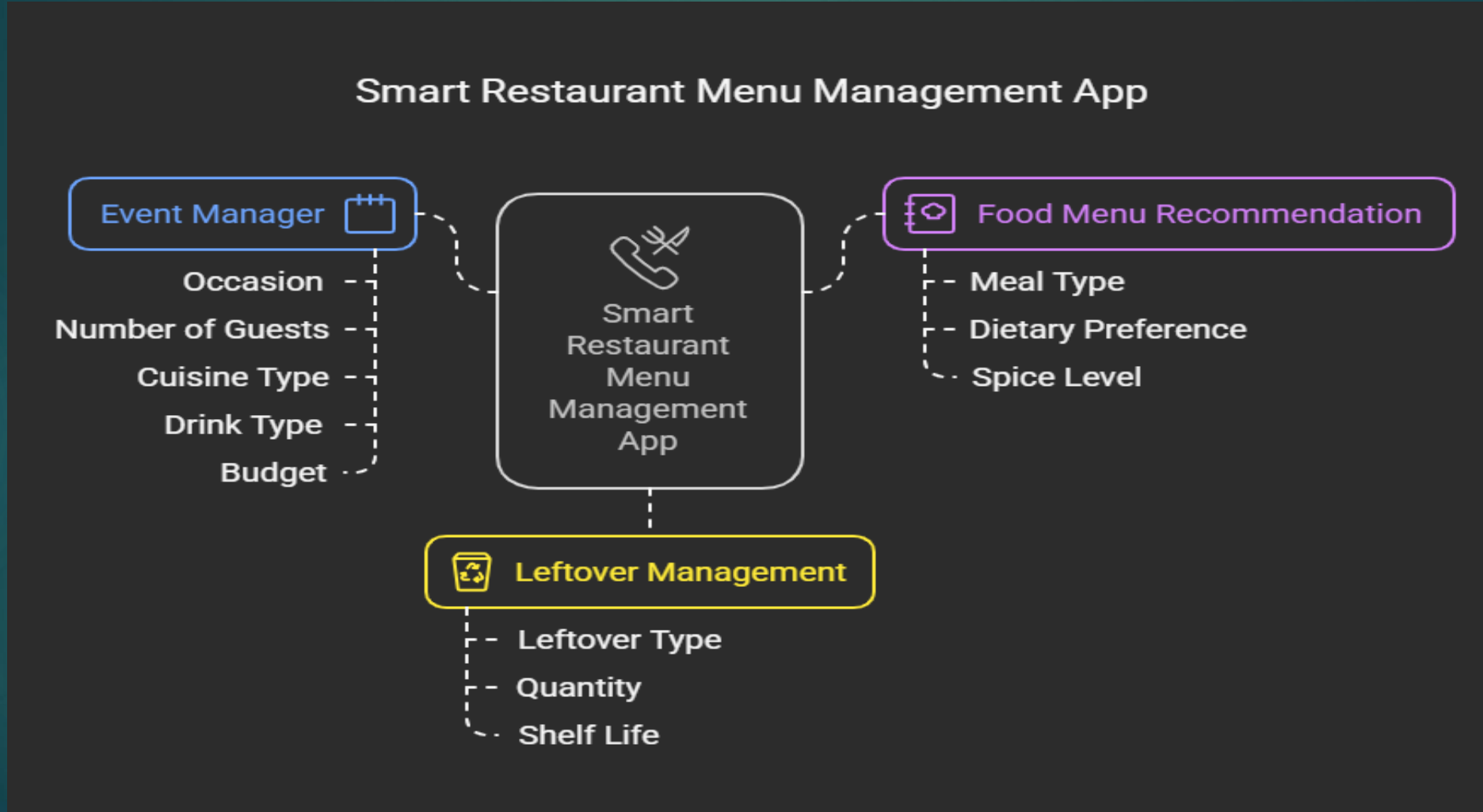
Gamification for Managers,Chef

Managers

It is Divided into three :

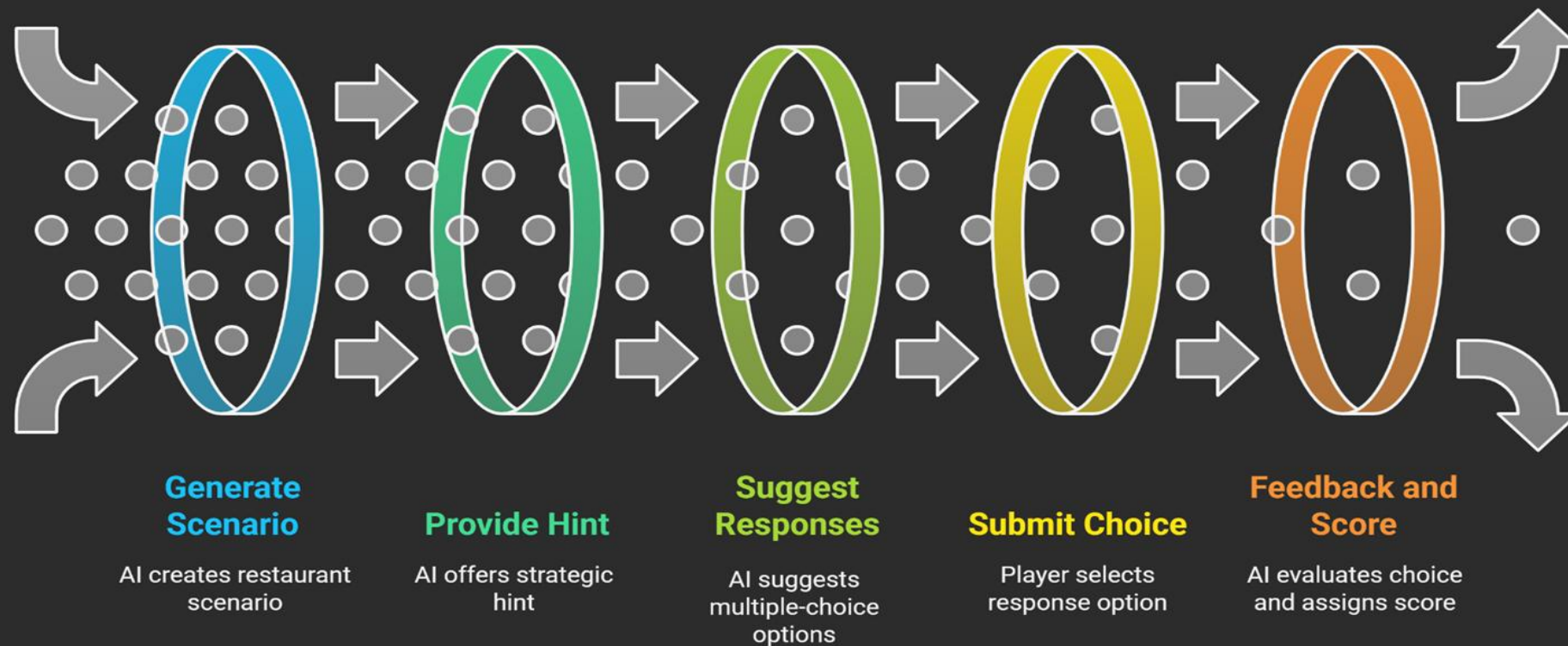
1. Event Manager
2. Food Menu Recommendation
3. Leftover Management
4. Gamification for Managers

Flowchart for Managers 1



Flowchart for Managers 2

AI-Powered Restaurant Challenge Process



Chef

1. Multimodels For Chef
2. Gamification for Chef

Flowchart for chef 1

Recipe Generation Process

Text Input
User enters preferences



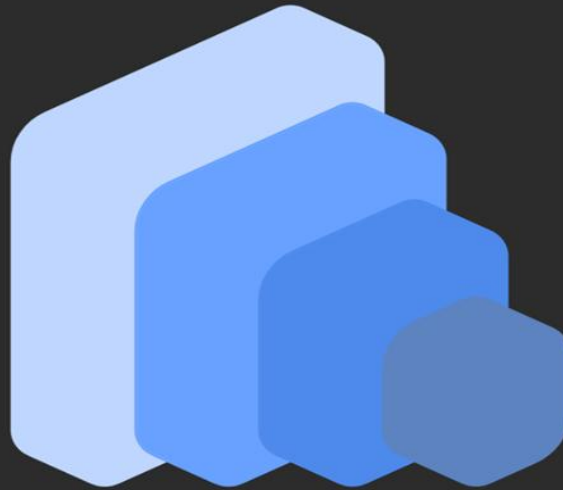
PDF Input
User uploads PDF



Image Input
User uploads image



Seasonal Recipe
User inputs seasonal data



Flowchart for chef 2

Top Recipe Contest Winners



Daily Winner

Daily top scorer in
recipe contest, latest
entries.



Weekly Winner

Weekly top scorer in
recipe contest,
consistent
performance.



Monthly Winner

Monthly top scorer
in recipe contest,
long-term
excellence.

APPS USED

- ▶ Bnanai.ai for user interface
- ▶ Grmini.si for code
- ▶ Streamlit
- ▶ Huggingface

Conclusion

- ▶ The Smart Restaurant Management System leverages cutting-edge Generative AI models to drive efficiency, reduce costs, and enhance customer experience. By integrating AI-powered personalization, automation, and predictive analytics, this solution addresses core industry challenges and sets a new benchmark in intelligent restaurant operations.

